

Saanvi Patel

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EDUCATION

B.S. Computer Science

Arizona State University, Tempe, AZ

August 2022 - May 2026

3.98 GPA

TECHNICAL SKILLS

Programming Languages: Java, Python, C++, JavaScript, C#, SQL, Swift, Bash

Framework & Technologies: .NET, SwiftUI, Firebase, Firestore, JUnit, Selenium

Developer Tools & Platforms: Git, Visual Studio, Jira, Slack, Windows, macOS, Linux, Ubuntu

WORK EXPERIENCE

Software Development Intern, GlaxoSmithKline (GSK) - Durham, NC

May 2025 - August 2025

- Engineered a unified monitoring platform in C# (WinForms) that consolidated 12 fragmented compliance tools into a single scalable dashboard, reducing manual workflows by 70%.
- Automated jurisdictional checks and employee record validations, cutting weekly analyst effort by 12+ hours and improving regulatory response time by 50%.
- Designed and implemented a dynamic SQL query engine with caching and query optimization, reducing query execution overhead by ~35% and enabling scalable onboarding of future monitors without code duplication.
- Developed self-validating UI logic with manual and auto-check using SQL queries, role-based branching, and embedded workflow triggers, eliminating 80% of repetitive analyst steps across the compliance process.

Administrative Assistant, Office of Research & Sponsored Projects Administration - ASU

May 2023 - May 2026

- Partnered with post-award services teams to administer \$70M+ in research funds, ensuring 100% compliance with funding protocols and improving financial transparency.
- Developed Python automation scripts and SQL-based financial reconciliation tools, reducing manual workload and financial record-keeping errors by 25%, improving cross-departmental coordination by 20%.
- Streamlined financial data management by reconciling 1,000+ transactions across ASU Foundation, Workday, and ERA systems, enhancing reporting accuracy.
- Mitigated discrepancies involving \$1.8M by issuing 200+ invoices, scheduling installments, and processing 500+ checks and wire transfers efficiently.

PROJECTS

CPQE - AI Educational Resource Recommendation Engine (*Python, AI Systems, Data Processing*)

March 2026

- Built AI-powered platform that generates personalized educational resource recommendations for bilingual families.
- Developed backend pipelines to process user queries, filter curated datasets, and rank relevant learning resources.
- Implemented intelligent recommendation logic to match student needs with contextual educational content.

Real-Time Event Notification System (*Java, WebSockets, REST APIs*)

August 2025

- Engineered a backend service to deliver real-time notifications to subscribed users using WebSocket connections.
- Designed REST APIs for event publishing and user subscription management.
- Implemented caching to support concurrent event delivery and reduce notification latency.

ASU Chirp - Social Media App (*SwiftUI, Firebase, Firestore, MVVM*)

May 2025

- Developed a SwiftUI app for students to share posts, events, and updates with real-time Firestore synchronization.
- Integrated Firebase Authentication, push notifications, and Firestore database synchronization.
- Streamlined state updates and improved UI responsiveness by 40% using MVVM architecture.

Heart Health Imaging & Monitoring System (*Java, JavaFX, UML Design*)

August 2024

- Developed a patient record system with UML use-case diagrams, CRC models, and class diagrams.
- Designed a JavaFX-based UI, reducing data entry errors by 25% in CT scan tracking workflow.
- Engineered secure patient data retrieval functionalities and optimized database access for medical records.

EXTRACURRICULAR & VOLUNTEER EXPERIENCE

Rewriting the Code - National Community Supporting Women in Tech

November 2024 - Present

CodePath: Technical Interview Prep - Advanced

May 2025 - Aug 2025

American Association for the Advancement of Science (AAAS)

January 2025 - May 2026

Marketing Coordinator, Connecting International Communities

May 2023 - Dec 2024